to read a map, and the other allows you to pan and zoom around your map. Both of these follow the same format and functionality of other behaviors and generators in D3, but are particularly useful for maps.

7.2.1 Graticule

A *graticule* is a grid line on a map. Just as D3 has generators for lines, areas, and arcs, it has a generator for graticules to make your maps more beautiful. The graticule generator creates gridlines (you can specify where and how many, or use the default) and also creates an outline that can provide a useful border. Listing 7.9 shows how to draw a graticule beneath the countries we've already drawn. Instead of `.data` we use `.datum`, which is a convenience function that allows us to bind a single datapoint to a selection so it doesn't need to be in an array. In other words, `.datum(yourDatapoint)` is the same as `.data([yourDatapoint])`.

Listing 7.9 Adding a graticule

```
var graticule = d3.geo.graticule();

d3.select("svg").append("path")
   .datum(graticule)
   .attr("class", "graticule line")
   .attr("d", geoPath)
   .style("fill", "none")
   .style("stroke", "lightgray")
   .style("stroke-width", "1px");

d3.select("svg").append("path")
   .datum(graticule.outline)
   .attr("class", "graticule outline")
   .attr("d", geoPath)
   .style("fill", "none")
   .style("stroke", "black")
   .style("stroke-width", "1px");
```

But how are we drawing so many graticule lines in figure 7.8 from a single datapoint? The `geo.graticule` function creates a feature known as a *multilinestring*. A multilinestring, as you may have figured out, is an array of arrays of coordinates, each corresponding to separate individual components of a feature. Multilinestrings and their counterparts, multipolygons, have always been a part of GIS because countries like the United States or Indonesia are made up of disconnected features such as states and regions, and that information needed to be stored in the data. As a result, when `d3.geo.path` gets a multipolygon or multilinestring, it draws a `<path>` element made up of multiple, disconnected pieces.

7.2.2 Zoom

You dealt with zoom a little bit in chapter 5, when you saw how the zoom behavior can easily allow you to pan a chart around the screen. Now it's time you start zooming with zoom. When we first looked at the zoom behavior, we used it to adjust the `transform`